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Approaches, challenges and prospects for modeling macroalgal dynamics in the green tide: The case of *Ulva prolifera*

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ABSTRACT

Ulva prolifera blooms, also known as green tides, significantly disrupt coastal ecosystems by altering species balance, energy flow, and nutrient cycling. These blooms, driven by nutrient enrichment, climate change, and human activities, have become a pressing environmental challenge in coastal regions. This review synthesizes current advances in modeling the growth, dispersal, and decline of *U. prolifera* blooms, emphasizing the roles of environmental drivers such as nutrient availability, temperature, light, and hydrodynamic conditions. We discuss empirical and mechanistic modeling approaches, highlighting their applications, limitations, and potential for predicting bloom dynamics. Special attention is given to model calibration, validation, and the integration of remote sensing and environmental data, which are crucial for ensuring model accuracy and reliability. Despite significant progress, challenges remain in addressing data gaps, incorporating climate variability, and simulating complex ecological interactions. Future research directions include the development of multi-scale, coupled models and the integration of socio-economic impacts to enhance bloom management strategies and inform policy development. The insights presented are intended to advance the understanding of *U. prolifera* bloom dynamics and contribute to the mitigation of their ecological and socio-economic impacts.

1. Introduction

Ulva prolifera, a widely distributed green macroalga, has become a focal point of research due to its rapid growth and frequent occurrence of large-scale blooms, particularly in eutrophic coastal waters (Fig. 1) (Zhang et al., 2015; Han et al., 2022; Cao et al., 2023; Yang et al., 2023b). These blooms, often referred to as green tides can have profound ecological and economic consequences, including hypoxia, changes in nutrient dynamics, and adverse effects on local fisheries, tourism, and water quality (Son et al., 2015; Xu et al., 2016). The growth, decline, and dispersal of U. prolifera are regulated by a complex interplay of biological and environmental factors. In addition to hydrodynamic forces and temperature, nutrient uptake efficiency plays a crucial role in determining biomass accumulation (Fan et al., 2023; Wang et al., 2023). The ability of *U. prolifera* to rapidly absorb and utilize nitrogen and phosphorus influences bloom formation, persistence, and decline(Wang et al., 2022a). As coastal ecosystems face increasing pressures from anthropogenic activities and climate change, predicting and managing *U. prolifera* blooms has become more critical than ever (Chen et al., 2020).

Mathematical and computational modeling has proven to be an invaluable tool in simulating and forecasting the behavior of *U. prolifera* blooms (Xu et al., 2016; Xu et al., 2023). Models of *U. prolifera* growth and dispersal are diverse, ranging from empirical models that rely on observational data to more mechanistic models that incorporate biological and environmental interactions (Sun et al., 2020b; Wang et al., 2022a). These models enable the prediction of bloom dynamics, including the onset, peak, and decline of biomass, under varying environmental scenarios (Kim et al., 2019; Sun et al., 2020b). However, despite advances in modeling techniques, many challenges remain in accurately capturing the nonlinear, dynamic nature of *U. prolifera* bloom processes, especially in the face of incomplete data and the variability of environmental conditions (Choi et al., 2023; Xu et al., 2023).

This study aims to provide a comprehensive synthesis of the current

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modeling approaches for *U. prolifera*, with a focus on the mechanisms driving growth, dispersal, and decline. We will examine the strengths and limitations of different modeling strategies, from simple empirical correlations to complex, process-based models, and highlight their applications in predicting *U. prolifera* bloom events. Special attention will be given to the major uncertainties that continue to affect model accuracy, including spatial and temporal scales, data gaps, and the role of unaccounted-for environmental variables. In doing so, we aim to identify key areas where further research and methodological advancements are needed, ultimately contributing to the development of more reliable and actionable predictive tools for managing *U. prolifera* blooms.

2. Materials and methods

2.1. Data sources

For the systematic data search, we accessed the Web of Science database (https://webofscience.clarivate.cn/) with publications on November 11, 2024. The time frame for our literature search spanned from 1980 to 2024 to ensure comprehensive coverage of scientific literature in the database. To compile the list of plant genera from selected studies, we cross-referenced with synonyms according to https://www.marinespecies.org/, ultimately using only the currently accepted names in our analyses (e.g., *Ulva* is the currently accepted name for *prolifera*).

The search combination used was: (Topic = "*Ulva prolifera*" OR "*Enteromorpha prolifera*" OR "*U. prolifera*" OR "*E. prolifera*" OR "green tide" OR "algal bloom") AND (Topic = "model*") AND (Topic = "sea"). The literature retrieval yielded 738 references from the Web of Science. After applying criteria to filter articles using the Web of Science core collection and Chinese Science Citation Database, the count was $N = 675$. After eliminating duplicate entries and screening entries based on titles, abstracts, and keywords to exclude records unrelated to numerical models, further screening was conducted to include only articles with clear images or data, the final count was $N = 41$. Endnote, an open-source bibliographic manager, was used for storing and handling the final database (https://endnote.com/), resulting in a pool of 41 records for meta-data extraction and analysis. To enhance the discussion, additional relevant literature, such as the history of numerical models and their applications in other fields, was also reviewed.

2.2. Literature review

The number of publications and frequency of citations related to

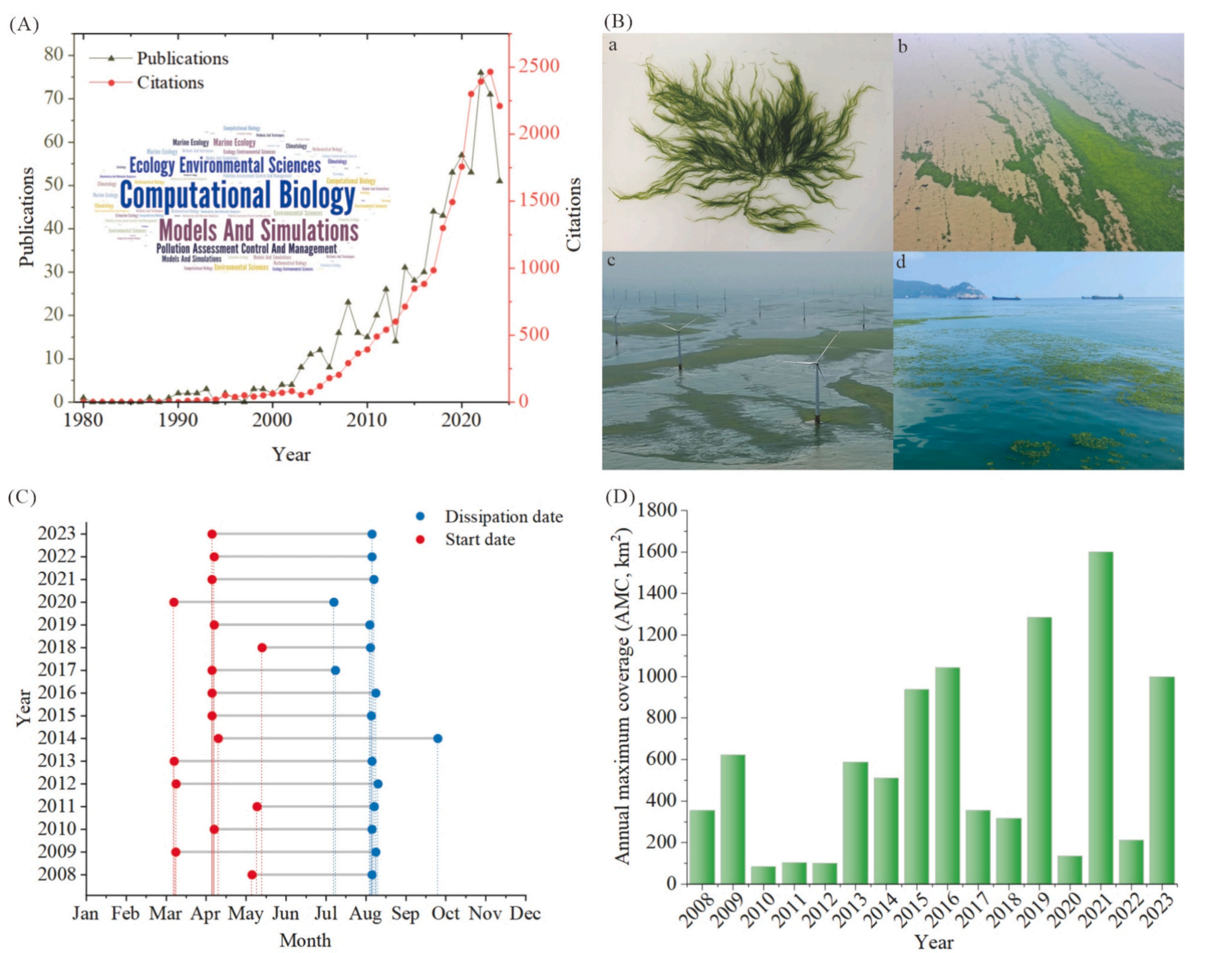


Fig. 1. The number of publications and citation frequency related to the model of *U. prolifera* from 1980 to 2024 along with a word cloud providing the most frequently used keywords in our database (A); Morphological characteristic of free-floating *Ulva* species (Ba), massive *U. prolifera* (green tide) blooms in the northern Yellow Sea off the coast of Jiangsu (Bb,c,d); Start and dissipation times (C) and annual maximum coverage (AMC) (D) of *U. prolifera* event in the Yellow Sea in different periods, based on data from the China Marine Disaster Bulletin, the Jiangsu Provincial Marine Disaster Bulletin, Cao et al. (2023), and Xu et al. (2023). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

U. prolifera is presented in Fig. 1A, illustrating a significant upward trend in both metrics from 1980 to 2024. Notably, the highest peaks in both publication count and citation frequency have occurred in the past five years, indicating a surge in scholarly interest in *U. prolifera*. This upward trajectory reflects a growing recognition of *U. prolifera* as a critical environmental issue, underscoring its importance in the fields of marine ecology, environmental science, and pollution management.

According to the analysis conducted using the Web of Science, the H-index for *U. prolifera* research stands at 70, indicating a high level of scholarly impact and productivity. The average citation count per article is 31.91, with a total of 21,190 citations recorded. After removing self-citations, the number of unique references cited amounts to 20,140, which further demonstrates the broad academic engagement with the topic. Additionally, key journals suitable for publishing important research on U. prolifera include Harmful Algae and Marine Pollution Bulletin. The primary contributing institutions are the Chinese Academy of Sciences and Ocean University of China. Detailed information can be found in Tables S1 and S2 (provided in the supplementary materials).

Further analysis of the key concepts in the *U. prolifera* literature reveals that computational biology is the most frequently mentioned area of research, reflecting the increasing use of advanced computational models and simulations to study *U. prolifera* dynamics. This is followed by significant mentions of models and simulations, indicating the central role of numerical approaches in understanding and predicting the behavior of *U. prolifera*. Additionally, ecology and environmental sciences, pollution assessment and control, and marine ecology are prominently featured, highlighting the interdisciplinary nature of research on *U. prolifera* and its relevance to global environmental concerns (Fig. 1A).

3. Modeling framework

3.1. Model development and theory

Modeling the dynamics and spread of *U. prolifera* involves integrating biological, physical, and chemical processes into numerical models to simulate growth, dispersal, and interactions with environmental factors (Liu et al., 2023; Yang et al., 2023b). A conceptual framework for such modeling includes selecting appropriate model types, incorporating key environmental drivers, conducting rigorous calibration and validation, and applying results to guide management and mitigation strategies (Fig. 2) (Zhou et al., 2021; Wang et al., 2022a; Ji et al., 2023; Wang et al., 2023). The integration of multiple model types, such as hydrodynamic, biogeochemical, and ecological models, and the use of sensitivity analyses are essential for improving predictive accuracy and supporting ecosystem management (Fan et al., 2023; Xu et al., 2023).

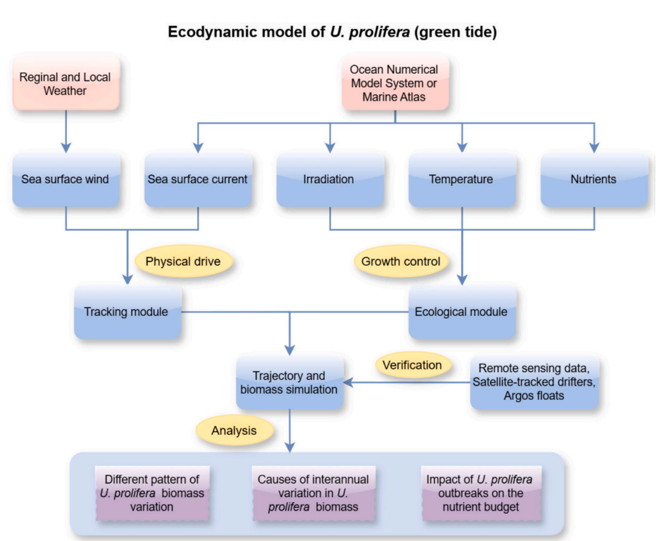


Fig. 2. The framework of ecological dynamics model of *U. prolifera* (green tide). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The development of numerical models often involves simplifying assumptions to balance computational feasibility and realism (Yu et al., 2023). These include assuming homogeneous environmental conditions, linear growth rates, and constant biological parameters, despite real-world variability in nutrient availability, hydrodynamics, and ecological interactions (Perrot et al., 2007; Kazamia et al., 2016; Zhou et al., 2021; Cao et al., 2023; Pan et al., 2023; Wu et al., 2024). Models also frequently rely on idealized hydrodynamic conditions and exclude small-scale variability, such as local turbulence or episodic weather events, which can significantly influence *U. prolifera* dynamics (Chojnacka, 2008; Wang et al., 2012a). In addition to physical and chemical drivers, biological interactions play a crucial role in *U. prolifera* bloom dynamics. Bacteria associated with *U. prolifera* can enhance algal growth by facilitating nutrient cycling, while some bacterial species secrete algicidal compounds that contribute to bloom termination (Zhang et al., 2021). Furthermore, viral infections can lead to sudden algal mortality, regulating bloom intensity and duration (Kim et al., 2020). However, many models disregard interactions with other species, oversimplifying the complexity of marine ecosystems (Li et al., 2014a; Magnusson et al., 2016). Recognizing these limitations is critical for refining model accuracy and ensuring reliable application in research and management. Future research should focus on coupling physical-biological models with microbial dynamics to improve green tide predictions.

The accuracy and applicability of *U. prolifera* models are often constrained by uncertainties in input data and biological parameters, particularly in regions with sparse monitoring infrastructure (Gao et al., 2021; Zhou et al., 2021). These models frequently struggle to capture interactions across spatial and temporal scales and fail to fully address the complexity of marine ecosystems, including species interactions and ecological feedback. Climate change and environmental variability add further challenges, as shifting oceanographic conditions are difficult to predict accurately (Su et al., 2024). Insufficient real-time and historical data limit model validation and calibration, while anthropogenic impacts, such as nutrient pollution and coastal development, are often inadequately integrated, leading to gaps between model predictions and observed dynamics (Wang et al., 2019; Ennackal and Odaneth, 2024). Addressing these issues requires high-quality, high-resolution data and the incorporation of nonlinear processes to better reflect algal behavior.

Despite these challenges, numerical models provide valuable insights into the dynamics of *U. prolifera*. By coupling processes such as nutrient cycling, hydrodynamic transport, and ecological interactions, these models can simulate the complex interplay of factors influencing *U. prolifera* bloom formation and dispersal (Zhou et al., 2021; Wang et al., 2022a; Fan et al., 2023). Enhancements such as coupled multi-scale models and regular validation against real-world observations can improve predictive reliability, particularly under changing environmental conditions (Xing et al., 2023; Jiang et al., 2024).

Understanding the environmental drivers of *U. prolifera* dynamics is crucial for accurate predictive modeling. Key factors include temperature, which regulates metabolic processes and can induce stress under extreme conditions, and nutrient concentrations, especially nitrogen and phosphorus, which are critical for growth (Wang et al., 2021; Yang et al., 2023a). Hydrodynamic forces, such as currents and vertical mixing, influence dispersal and nutrient availability, while light availability, affected by turbidity and seasonal changes, drives photosynthesis (Angles et al., 2008). Other factors, such as salinity, ocean acidification, and storm events, further shape *U. prolifera* bloom dynamics by affecting algal health, carbon fixation, and nutrient redistribution (Dudgeon and Kubler, 2020; Sanchez-Arcos et al., 2022). Comprehensive models that integrate these drivers are essential for predicting bloom behavior and guiding effective management strategies.

The mathematical modeling of *U. prolifera* dynamics typically involves formulating growth, decline, and dispersal processes through various equations. Growth models often assume exponential or logistic growth, with nutrient availability serving as a limiting factor (Li et al., 2018b; Xu et al., 2023). Mortality is modeled using linear or density-dependent processes, reflecting the effects of environmental stress, predation, and competition (Wang et al., 2020; Su et al., 2024). Dispersal is driven by hydrodynamic forces and modeled through advection-diffusion equations that account for water currents and turbulence (Obolski et al., 2022). Nutrient uptake efficiency is a key determinant of U. prolifera bloom biomass. Empirical studies have shown that the maximum uptake rate of dissolved inorganic nitrogen (DIN) and phosphorus (DIP) varies significantly across bloom stages (Sun et al., 2020a). A more refined parameterization of nutrient kinetics, including half-saturation coefficients (KDIN, KDIP) and maximum uptake rates (VmDIN, VmDIP), should be incorporated into mechanistic models to improve biomass predictions. Furthermore, dynamic stoichiometric modeling approaches that integrate cellular nitrogen and phosphorus quotas can enhance the accuracy of bloom growth simulations. Coupled models integrating these processes provide a holistic view of *U. prolifera* bloom dynamics, enabling accurate predictions under diverse environmental conditions and supporting efforts to mitigate the impacts of green tides.

3.2. Remote sensing and environmental data collection

Remote sensing and environmental data collection are indispensable for studying the dynamics of marine ecosystems, particularly in understanding the growth, decline, and dispersal of *U. prolifera*. Remote sensing tools, including satellites and drones, provide comprehensive coverage of algal biomass, water quality, and environmental conditions, while in-situ sensors deliver localized, high-frequency measurements that enhance data precision (Xiao et al., 2017; Xu et al., 2023; Peter et al., 2024). Additionally, coupling hydrodynamic models with satellite-derived data has facilitated the tracking of green tide movement and growth, improving the spatial and temporal resolution of predictions. Techniques like the maximum cross-correlation method have refined model accuracy, helping to identify key environmental drivers of *U. prolifera* bloom dispersal and persistence(Ji et al., 2023). Satellites equipped with sensors like MODIS and Sentinel-2 generate imagery that detects and tracks *U. prolifera* blooms, estimates biomass concentrations, and monitors chlorophyll-a levels, a key indicator of *U. prolifera* presence (Table 1) (Kim et al., 2019; He et al., 2021; Cao et al., 2023; Xu et al., 2023). Hyperspectral sensors further refine detection by offering detailed spectral information, improving the accuracy of biomass estimates, and enabling the identification of *U. prolifera*. In addition to satellite-based tools, airborne sensors and drones facilitate high-resolution data collection in localized coastal regions (Son et al., 2015; Peter et al., 2024). Drones provide real-time imagery and detailed insights into *U. prolifera* concentrations, water quality, and environmental conditions, addressing gaps in satellite coverage. This combination of remote sensing platforms enhances the ability to monitor *U. prolifera* dynamics across diverse spatial scales.

In-situ data collection complements remote sensing by offering continuous, high-precision measurements of environmental variables such as temperature, salinity, nutrient concentrations, and oxygen levels (Table 2) (Chen et al., 2020). These data are gathered using oceanographic sensors deployed on buoys, ships, and underwater platforms, which provide essential inputs for understanding *U. prolifera* growth and bloom dynamics (Bao et al., 2015a; Tang et al., 2024). Integrating remote sensing and in-situ data into predictive models allows researchers to simulate *U. prolifera* dynamics more effectively. These models incorporate environmental drivers such as nutrient levels, water temperature, and light availability to predict algal growth, decline, and

Table 1
Information on the satellites used for monitoring and predicting *U. prolifera.*

Sensor	Terra/Aqua MODIS	Landsat-5/7/8 TM/ETM/OLI	HJ-1 CCD	GF-1/6 WFV	Sentinel-2 MSI	HY-1 CZI	HJ-2 CCD	GF-3 SAR	Sentinel-1 SAR	COMS GOCI	Radarsat1/2 C-SAR
Country or region	US	US	China	China	ESA	China	China	China	ESA	Korea	Cannada
Data Availability	1999-	1984-	2008-	2013-	2016-	2018-	2020-	2016-	2014-	2010–2021	1996–2013
Spatial resolution (m)	250,500	30	30	16	10,20,60	50	16	1–500	20	500	30–100
Revist period (days)	1	16	4	4	10	3	4	1.5	12	1/3	24
Swath(km)	2330	180	700	800	290	950	800	10–650	250	2500	18–500
Acquisition modes	n/a	n/a	n/a	n/a	n/a	n/a	n/a	Wide fine stripmap, Standard stripmap …	Interferometric wide …	n/a	Standard beam, wide beam…
Polarization type	n/a	n/a	n/a	n/a	n/a	n/a	n/a	VH + VV …	VH + VV …	n/a	VH + VV …
Usage	GMB monitoring in the whole Yellow Sea	GMB monitoring in the source region (Subei shoal) in the early stage of the bloom (Sentinel-2 MSI images are also used to extract the Porphyra mariculture)									

NOTE: Landsat-5 orbit operation time (March 1984–June 2013).

Table 2
Environmental element data sources, modified from Chen et al. (2020).

Data	Calculation formula	Source
Pollutants on the Jiangsu coast	/	National Marine Environmental Monitoring Center (http://www.soa.gov.cn/) Jiangsu Environmental Monitoring Center (http://www.jsem.net.cn/)
Rivers discharge	/	Jiangsu Province Hydrology and Water Resources Investigation Bureau (http://www.jsswj.com.cn/) Nantong Marine Environment Monitoring Center (http://www.ntjczxz.com/)
Direct discharging and land-based pollution	/	Department of Ecology and Environment of Jiangsu Province (http://hbt.jiangsu.gov.cn/)
Mariculture	FAqua = MAqua × AAqua (where AAqua is marine culture area, MAqua is pollutant emissions in unit culture area)	Jiangsu Province Statistical Bureau (http://tj.jiangsu.gov.cn/index.html)
Surface runoff outside embankments	F = Q·C·Adiwai (where Adiwai is the land area between sea embankment and the average high tide line on the Jiangsu coast, C is the concentration of pollutants in rainwater, and Q is the rainfall.)	National Marine Environment Monitoring Center (https://www.nmemc.org.cn/) China Meteorological Administration (http://www.cma.gov.cn/)
Atmospheric wet deposition	F = Q·C·A (where A is the study area, C is the concentration of pollutants in rainwater, and Q is the rainfall.)	China Meteorological Administration (http://www.cma.gov.cn/)
Atmospheric dry deposition	F = V·Caero·A (where A is the study area, C is the aerosol concentration in the atmosphere, and V is the deposition rate.)	Jiangsu Meteorological Bureau (http://js.cma.gov.cn/)

dispersal (Yang et al., 2023a; Su et al., 2024). Remote sensing data plays a crucial role in calibrating and validating these models, enhancing their accuracy by aligning predictions with satellite observations. This integration not only facilitates the forecasting of *U. prolifera* bloom timing and locations but also supports early warning systems and long-term monitoring of bloom trends, such as changes in frequency, intensity, and spatial distribution (Glibert et al., 2014; Kang et al., 2019).

Despite their value, remote sensing and environmental data collection face several challenges that hinder their effectiveness. Satellite observations can be obstructed by cloud cover, limiting continuous monitoring in regions with frequent cloudiness. Additionally, the spatial resolution of satellite imagery may be insufficient for detailed studies in patchy coastal bloom areas. High-resolution data from drones or airborne sensors, while valuable, incur high costs and require specialized equipment. To address these challenges, rigorous quality control measures are essential. Calibration of remote sensing data with in-situ measurements ensures accurate biomass estimates, while field validation enhances reliability. Data cleaning processes, such as removing outliers and filling gaps, are necessary to maintain dataset integrity. Geospatial alignment of field and satellite data ensures spatial consistency, while temporal consistency and uncertainty quantification account for seasonal variability and sensor inaccuracies (Li et al., 2018a; Reimer et al., 2022). By integrating high-quality remote sensing and in-situ data with advanced modeling techniques, researchers can improve the predictive accuracy of *U. prolifera* bloom models. Enhanced data acquisition and quality control provide a comprehensive understanding of environmental drivers, enabling more precise simulations of bloom dynamics. This integrated approach is vital for developing effective coastal management and conservation strategies, ultimately mitigating the ecological and socio-economic impacts of *U. prolifera* blooms.

4. Numerical model implementation

4.1. Model types and key parameters

The choice of model type depends on the study's objectives and the availability of environmental data, with calibration and validation processes ensuring predictive accuracy for applications in scenarios such as nutrient loading, hydrodynamic changes, and climate impacts (Table 3). Hydrodynamic Models focus on the physical movement of water, capturing the influence of currents, tides, and wave action on *U. prolifera* dispersal (Fig. 3) (Bao et al., 2015b; Gao et al., 2021). These models are essential for understanding how *U. prolifera* fragments are transported across coastal and offshore regions. For instance, a triple-level mesh scheme described by Fan et al. (2023) integrates hydrodynamic processes at varying resolutions to capture both large-scale circulation patterns and small-scale nearshore dynamics. At the broadest scale, the Bohai Model employs 6403 elements with a resolution of 1200 m, while the Qinhuangdao Model refines this to 34,071 elements with a 10 m resolution, and the Jinmeng Bay Model achieves 4 m resolution with 25,503 elements, enabling precise simulations of complex multi-scale processes.

Research has shown that optimal growth conditions for *U. prolifera* have been identified as temperatures ranging from 14 °C to 27 °C and salinity levels between 26 and 32 (Gao et al., 2021). Deviations from these ranges often induce stress, reducing growth efficiency and potentially accelerating the decline phase of blooms. Numerical models for *U. prolifera* must incorporate these critical parameters to accurately simulate growth and decline dynamics. The temporal variability of temperature and nutrient concentrations, as reflected in Table S3, emphasizes the need for dynamic input data that captures seasonal and interannual changes. Furthermore, integrating light availability and salinity into model frameworks ensures a comprehensive representation of environmental drivers (Wang et al., 2022b). To enhance predictive capabilities, future research should focus on refining the resolution of temperature and nutrient datasets and integrating real-time monitoring systems into model calibration processes.

Numerical models simulating the growth, dispersal, and decline of *U. prolifera* rely on ecological modules that incorporate key parameters representing nutrient uptake, photosynthesis, and mortality (Wang et al., 2012c; Liu et al., 2017; Sun et al., 2022). These parameters are crucial for accurately reflecting the physiological and environmental responses of *U. prolifera*, providing insights into its bloom dynamics under varying environmental conditions (Table 4) (Sun et al., 2020a; Zhou et al., 2021; Wang et al., 2022b). Nutrient availability is a primary driver of *U. prolifera* growth. The boundary condition settings described by Wang et al. (2023) highlight significant variability in nutrient loads at riverine inputs, with NH_4-N levels ranging from 0.10 to 1.05 mg/L and NO_3-N levels reaching up to 6.72 mg/L. These inputs shape nutrient transport and availability in coastal ecosystems. The ecological module uses nitrogen (N) and phosphorus (P) quotas to capture the variability in nutrient uptake, Nitrogen Quotas (QNmin and QNmax) reflect the ability of *U. prolifera* to thrive in nutrient-rich environments and adapt to fluctuating N concentrations; Phosphorus Quotas (QPmin and QPmax) indicating the role of P as a limiting nutrient in coastal waters. The maximum uptake rates of dissolved inorganic nitrogen (DIN) and phosphorus (DIP), denoted as VmDIN and VmDIP. Half-saturation coefficients KDIN and KDIP determine nutrient uptake efficiency, with lower values indicating higher affinity for nutrient absorption. The photosynthetic capacity and efficiency of *U. prolifera* drive its rapid growth during bloom formation: Maximum Photosynthetic Rate (Pmax), *U. prolifera* demonstrates high photosynthetic activity under optimal conditions (Xu et al., 2014); Photosynthetic Efficiency (α), *U. prolifera*

Table 3
comparison of different types of *U. prolifera* bloom models: their suitability, advantages, and disadvantages.

Model type	Advantages	Disadvantages	Suitability	References
ROMS (Regional Ocean Modeling Systems)	High-resolution coastal and regional circulation.	Computationally intensive, requires expertise.	Modeling bloom dispersion and nutrient-driven growth in coastal regions.	(Lee et al., 2011; Huang et al., 2014; Son et al., 2015; Hu et al., 2018; Sun et al., 2020b; He et al., 2021; Chen et al., 2022)
POM (Princeton Ocean Model)	Computationally efficient, widely used in coastal studies.	Less adaptable to complex topography.	Modeling large-scale water circulation and algal bloom transport.	(Qiao et al., 2011; Wu et al., 2011; Zhang et al., 2017; Chen et al., 2022)
FVCOM (Finite Volume Community Ocean Model)	High resolution for complex coastal environments.	Increased computational complexity.	Simulating algal dispersion in estuarine and nearshore areas.	(Hu et al., 2010; Xiangyang et al., 2011; Li et al., 2014b; Bao et al., 2015b; Zhang et al., 2015; Chen et al., 2020; Gao et al., 2021; Zhou et al., 2021)
GNOME (General NOAA Operational Modeling Environment)	Easy setup for operational forecasts.	Focuses on surface processes, limiting subsurface dynamics.	Predicting surface bloom trajectories, especially during storms.	(Xu et al., 2016)
MIKE21&3 (MIKE Flow model FM)	Versatile for hydrodynamics and ecological simulations.	Commercial license required.	Simulating nutrient dynamics and bloom impacts on water quality.	(Ji et al., 2018; Chen et al., 2022; Fan et al., 2023)
MASNUM (Key Laboratory of Marine Science and Numerical Modeling)	Optimized for Yellow Sea regional simulations.	Less adaptable outside Chinese regions.	Regional-scale bloom dynamics focusing on local environmental drivers.	(Zhao et al., 2018)
HYCOM (Global Hybrid Coordinate Ocean Model)	Global simulations with multiple coordinate systems.	Requires substantial computational power.	Understanding large-scale oceanic drivers like nutrient transport.	(Zhao et al., 2018; Choi et al., 2023)
ROMS-CoSiNE (Regional Ocean Modeling Systems- Carbon, Silicate and Nitrogen Ecosystem)	Includes detailed ecosystem modeling.	High data requirements, complex.	Forecasting nutrient-driven growth and ecological impacts of blooms.	(Wang et al., 2022a; Wang et al., 2022b)
ARIMA (AutoRegressive Integrated Moving Average)	Effective for time-series predictions.	Limited in capturing spatial dynamics and ecological interactions.	Predicting bloom timing based on historical trends.	(Liu et al., 2023)
Delft3D	Flexible for hydrodynamics and ecological simulations.	Requires expertise and computational resources.	Studying bloom triggers with hydrodynamics and sediment transport.	(Wang et al., 2023)

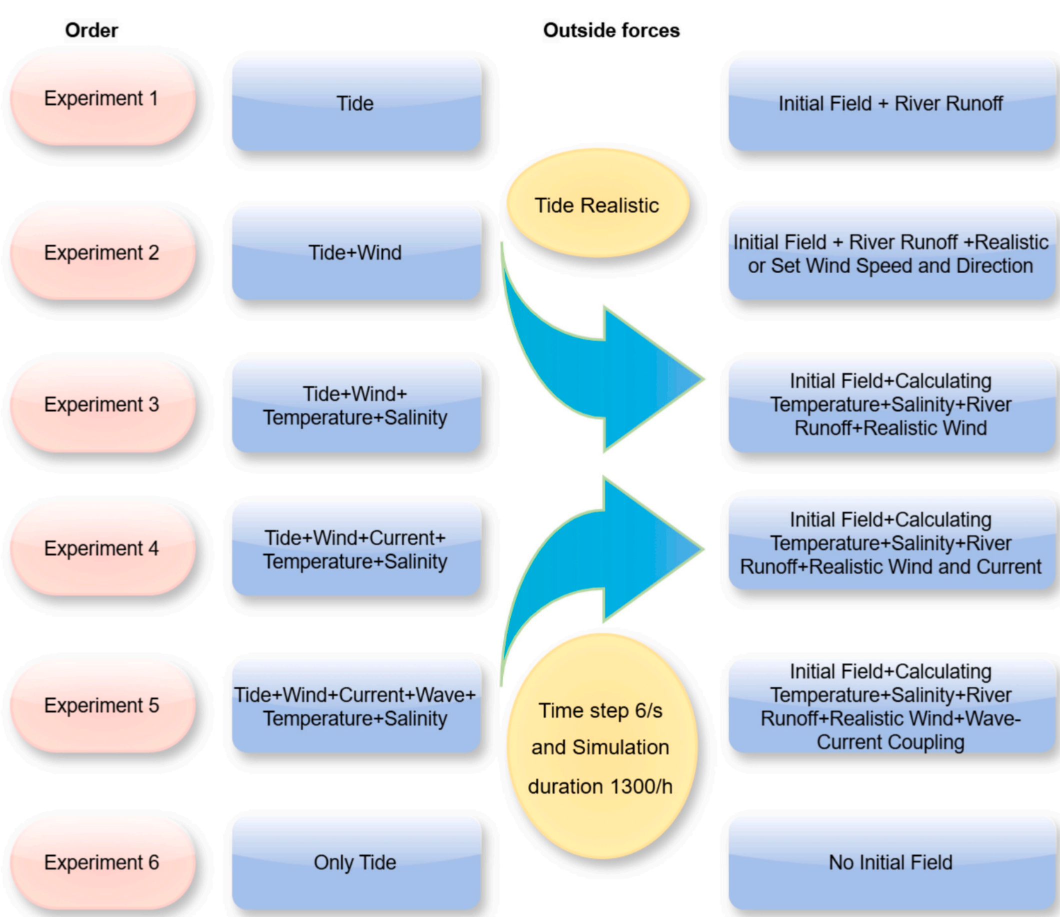


Fig. 3. The outside forces configuration for particle tracking experiments.

Table 4
Parameters used in the ecological module for *U. prolifera*.

Parameters	Description	Value	Dimension	Reference
QNmin	Minimum N quota	25.3	mmol N mol C^{-1}	(Fujita, 1985)
QNmax	Maximum N quota	108.7	mmol N mol C^{-1}	(Sun et al., 2015)
QPmin	Minimum P quota	0.097	mmol N mol C^{-1}	(Sfriso et al., 1990)
QPmax	Maximum P quota	1.4	mmol N mol C^{-1}	(Sun et al., 2015)
VmDIN	Maximum uptake rate of DIN	2.8	mmolNmolC^{-1} h^{-1}	(Li and Zhao, 2011; Luo et al., 2012)
VmDIP	Maximum uptake rate of DIP	0.58	mmolNmolC^{-1} h^{-1}	(Luo et al., 2012)
KDIN	Half-saturation coefficient for DIN	18.77	μmol N L^{-1}	(Li and Zhao, 2011; Luo et al., 2012)
KDIP	Half-saturation coefficient for DIP	10	μmol N L^{-1}	(Luo et al., 2012)
Pmax	Maximum photosynthetic rate	240.51	μmolCgWW^{-1} h^{-1}	(Xu et al., 2014)
α	Photosynthetic efficiency	2.52	(μmolCgWW^{-1} h^{-1})/ (μmolphotonsm-2 s^{-1})	(Xu et al., 2014)
Rd	Dark respiration rate	18.4	μmolCgWW^{-1} h^{-1}	(Xu et al., 2014)
θ_1	Growth coefficient	1.15	–	(Wang et al., 2022b)
θ_2	Growth coefficient	1.8	–	(Wang et al., 2022b)
θ_3	Mortality coefficient	1.1	–	(Wang et al., 2022b)
T_1	Maximum temperature for Ulva growth	15	°C	(Wang et al., 2022b)
T_2	Minimum temperature for Ulva growth	20	°C	(Wang et al., 2022b)
T_3	Critical temperature for mortality of Ulva	25	°C	(Wang et al., 2022b)
I_0	Optimal light for Ulva growth	5500	Lx	(Wang et al., 2022b)
Kn	Nitrate half-saturation coefficient	8.39	μmol/L	(Wang et al., 2022b)
Kp	Phosphorus half-saturation coefficient	0.904	μmol/L	(Wang et al., 2022b)
Gmax	Maximum growth rate	0.45	d^{-1}	(Wang et al., 2022b)
Dmax	Maximum mortality rate	0.1	d^{-1}	(Wang et al., 2022b)

efficiently converts light into energy; Respiration and Growth Coefficients: Dark respiration rate (Rd) is 18.4 μmol C g WW^{-1} h^{-1}, while growth coefficients (θ1, θ2) are 1.15 and 1.8, respectively (Wang et al., 2022b). Temperature Ranges, optimal growth occurs between 15 °C and 20 °C, with critical mortality thresholds above 25 °C (Wang et al., 2022b). Light Availability, optimal light intensity (I0) for growth is 5500 lx, ensuring sufficient energy for photosynthetic activity. The maximum growth rate (Gmax) and the maximum mortality rate (Dmax) encapsulate the rapid growth potential of *U. prolifera* under favorable conditions and its susceptibility to environmental stressors during bloom decline. The ecological parameters outlined above provide a comprehensive foundation for modeling *U. prolifera* dynamics. By incorporating these values into numerical models, researchers can simulate the nutrient-driven growth, light-dependent photosynthesis, and temperature-influenced mortality processes that characterize green tide events. Furthermore, understanding nutrient quotas and uptake kinetics highlights the critical role of nutrient management in mitigating bloom formation. The integration of precise photosynthetic and growth parameters ensures accurate predictions of bloom intensity and duration, while temperature and light thresholds inform models about the conditions leading to bloom cessation. Future research should aim to refine these parameters through high-resolution, field-based measurements across diverse coastal regions. Additionally, incorporating real-time environmental monitoring into model calibration processes can enhance the predictive accuracy of *U. prolifera* bloom models, contributing to more effective coastal management and mitigation strategies.

Incorporating hierarchical modeling frameworks, like the triple-level mesh approach, further enhances model precision by addressing spatial heterogeneity and multiscale interactions. Additionally, leveraging diverse environmental datasets and integrating boundary condition refinements allow for more accurate representation of complex marine ecosystems. Future advancements should focus on incorporating climate change projections, high-resolution data, and dynamic boundary conditions to improve the applicability of these models in diverse coastal and marine settings.

4.2. Model calibration and validation

Model calibration and validation are critical processes for ensuring the reliability and predictive accuracy of numerical models used to simulate the dynamics of *U. prolifera* (Ji et al., 2018). These processes involve aligning model outputs with observational data and testing the model's performance under real-world conditions (Liu et al., 2023). Effective calibration and validation enhance model applicability for forecasting *U. prolifera* bloom events and guiding management strategies. Calibration systematically adjusts model parameters to minimize discrepancies between simulated outputs and observed data. Key parameters for *U. prolifera* include nutrient uptake rates, growth coefficients, mortality rates, and hydrodynamic factors, which are fine-tuned using field observations and remote sensing data (Fig. 4–5) (Ji et al., 2018; Dudgeon and Kubler, 2020; Cao et al., 2023; Xu et al., 2023). Nutrient parameters, concentrations of nitrogen and phosphorus, along with their uptake rates, are calibrated using in-situ measurements and laboratory experiments to simulate *U. prolifera* growth under varying nutrient conditions accurately (Wang et al., 2023; Yang et al., 2023b). Temperature and Light dependencies, parameters governing temperature, and light thresholds are adjusted based on seasonal and spatial datasets, reflecting physiological responses of *U. prolifera* (Lin et al., 2022; Liu et al., 2023). Hydrodynamic inputs, currents, tidal forces, and wave action, critical for dispersal modeling, are calibrated using oceanographic sensor networks and satellite-derived hydrodynamic data (Xing et al., 2019). Advanced techniques, such as Bayesian inference and optimization algorithms, have increasingly been applied to refine parameter estimation (Feki-Sahnoun et al., 2017; Zhong et al., 2020). These methods provide probabilistic assessments of parameter uncertainties, thereby improving model robustness. One of the key challenges in biomass prediction is the time lag effect, which arises due to the decoupling between nutrient absorption and biomass increase during bloom drift. Field studies indicate that *U. prolifera* can drift 20–30 km per day under favorable wind and current conditions, meaning that the location of peak biomass accumulation may be spatially displaced from the original nutrient-enriched areas (Wang et al., 2023). This spatial and temporal decoupling introduces uncertainties in bloom modeling, particularly when calibrating growth parameters based solely on local nutrient availability. To address this issue, model calibration should integrate lagged growth response functions that account for the delayed impact of nutrient assimilation on biomass expansion. Incorporating internal nutrient storage dynamics (e.g., cellular nitrogen and phosphorus quotas) into bloom models can better capture the time-dependent relationship between nutrient uptake

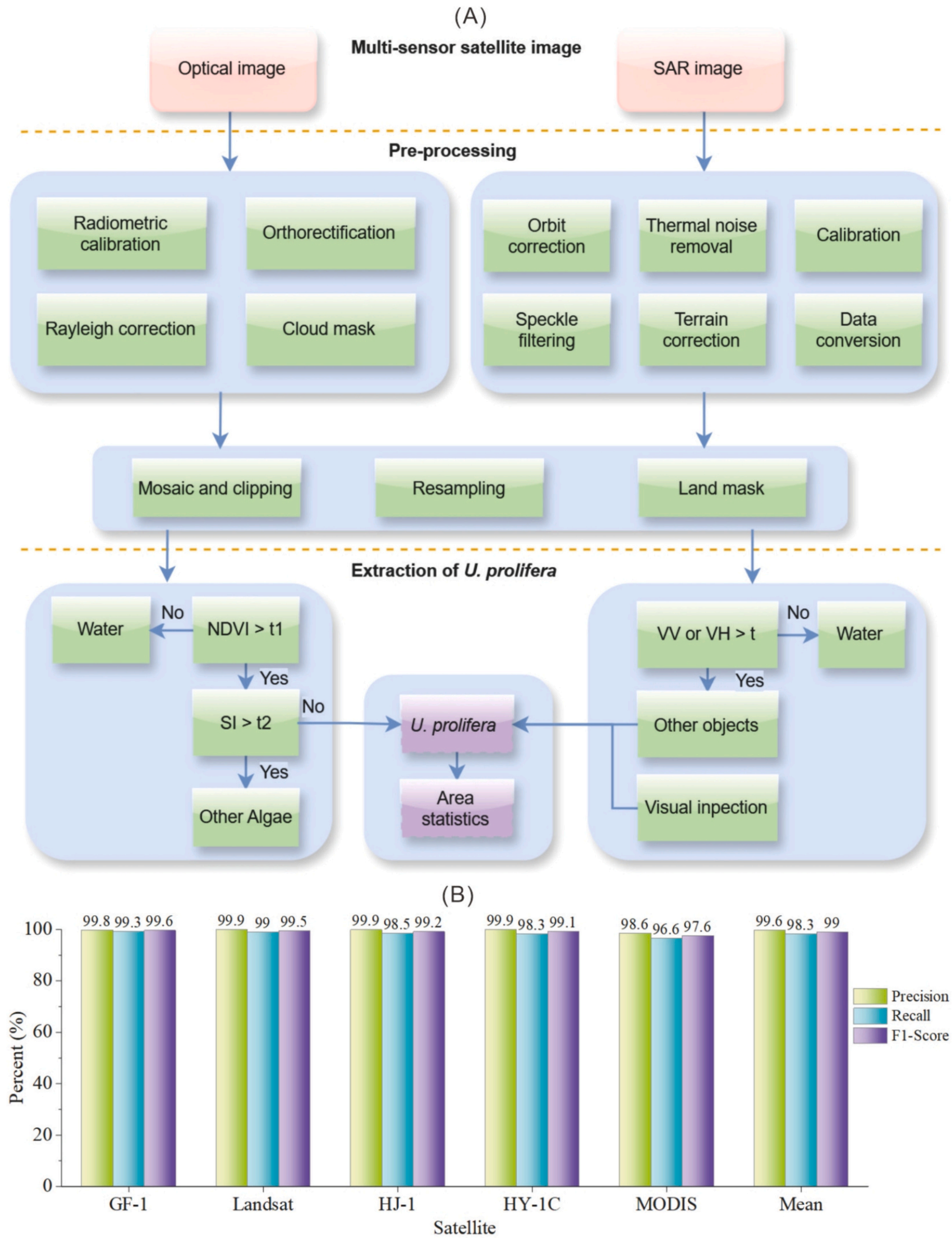


Fig. 4. Extraction of floating *U. prolifera* from satellite images (A), and accuracy evaluation of the extraction effect (B).

and subsequent growth (Sun et al., 2020a). Additionally, Lagrangian particle tracking models combined with biochemical growth modules can improve simulations by explicitly modeling the drift of *U. prolifera* while tracking internal nutrient reserves. Calibration of hydrodynamic and biological parameters should be carried out using multi-source validation approaches, remote sensing-based biomass estimations to determine the spatial distribution of bloom progression; buoy-based chlorophyll-a measurements to track in-situ bloom dynamics over time; in-situ nutrient profiling to assess uptake efficiency and its relationship with subsequent biomass increase. Sensitivity analysis suggests that growth coefficients (Gmax), nutrient uptake rates (VmDIN, VmDIP), and drift velocity (Vdrift) are key factors influencing biomass prediction accuracy. Future calibration efforts should incorporate machine learning techniques to optimize these parameters based on historical datasets and improve model predictive capabilities.

Validation evaluates the model's ability to simulate *U. prolifera* bloom dynamics under various environmental scenarios by comparing predictions against independent observational datasets. Common validation metrics include root mean square error (RMSE), coefficient of determination (R^2), and mean absolute error (MAE) (Zhang et al., 2021; Liu et al., 2023). Specific validation approaches include spatial validation, temporal validation, and scenario testing (Cao et al., 2023). Remote sensing imagery, particularly chlorophyll-a concentration maps, validates predictions of *U. prolifera*'s spatial distribution (Figs. 6–7) (Ji et al., 2018; He et al., 2021; Wang et al., 2022a). Time-series data from buoy networks and field surveys validate temporal dynamics, including

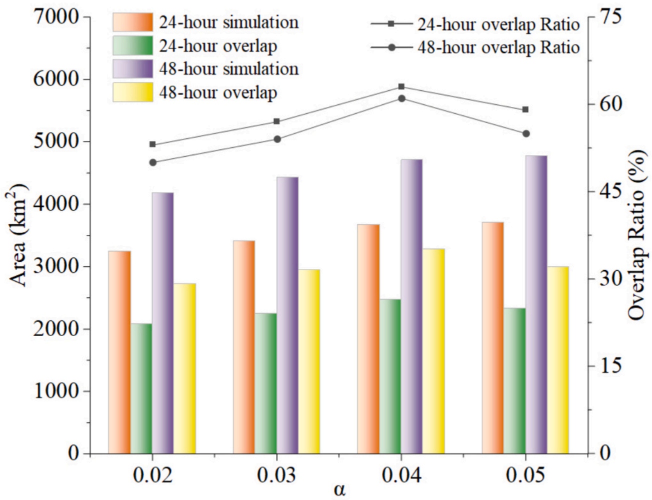
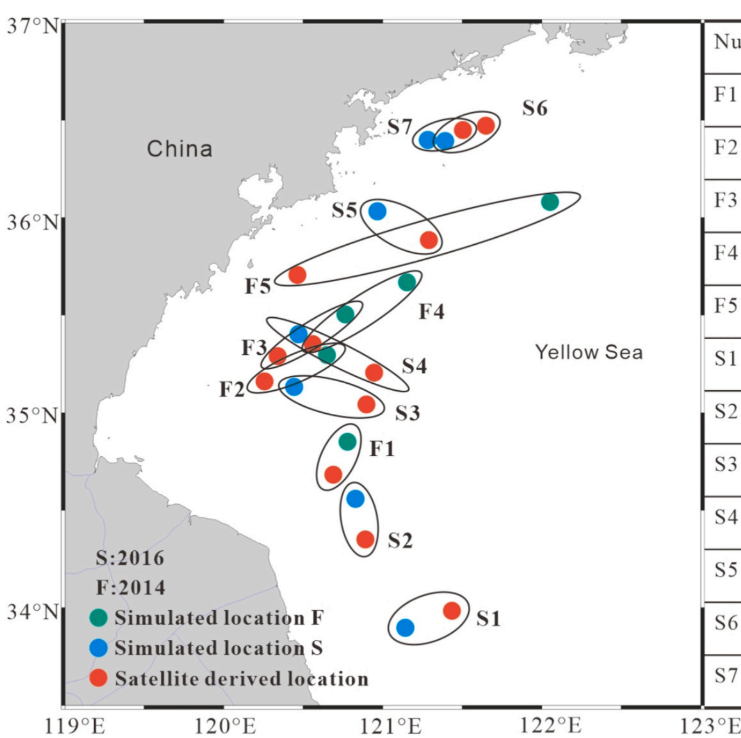


Fig. 5. Comparison of simulation results by different α values.

bloom onset, peak biomass, and decline phases (Fig.S1-S2, provided in the supplementary materials) (Zhao et al., 2018; Chen et al., 2020). Models are tested under varying nutrient loading, temperature, and hydrodynamic scenarios to assess predictive reliability across diverse conditions (Fig. 3) (Bao et al., 2015b; Gao et al., 2021). Integrated approaches combine calibration and validation to provide comprehensive insights into model performance. Iterative processes that integrate both steps enhance predictive accuracy over multiple cycles (Tsai et al., 2022; Yuan and Koropeckyj-Cox, 2022). Incorporating real-time data into models during calibration allows for dynamic updates to predictions, particularly during rapidly changing *U. prolifera* bloom events (Wang et al., 2012; Zhang et al., 2013).

Sensitivity analysis identifies influential parameters by systematically varying inputs, such as growth rates and nutrient concentrations, and observing their effects on model outputs (Han et al., 2022). Meanwhile, uncertainty assessment addresses potential errors arising from data inaccuracies, model assumptions, and parameter uncertainties. Techniques such as Monte Carlo simulations and Bayesian methods assess these uncertainties by exploring a range of possible outcomes and providing confidence intervals for predictions (Golubkov et al., 2018; Yoshioka and Hamagami, 2024). The relative errors between observed data and simulated results presented in Fan et al. (2023) indicate variability in the model's accuracy across different sites and components, with relative errors for chemical oxygen demand (COD) ranging from 8.20 % to 33.50 %, and for nitrate concentrations from 3.50 % to 19.70 %. These variations suggest that while the model performs well in some scenarios, further refinement may be required to improve accuracy in regions with higher discrepancies. These results highlight the need for site-specific calibration. Tidal model evaluation showed significant discrepancies for the M2 tidal constituent at some locations, such as Bayuquan (17.16 cm) and Changxingdao (13.80 cm), underscoring the importance of localized refinements (Han et al., 2022). Hydrodynamic model assessments achieved high skill scores (0.92 to 0.99) for tidal-level simulations but indicated moderate performance for wave and current predictions, suggesting areas for improvement (Wang et al., 2023). The calculated mean floating speed of *U. prolifera*, as reported by Son et al. (2015), varied significantly across three days in July 2011 based on hourly daytime measurements. These variations highlight the dynamic nature of *U. prolifera* movement, likely influenced by local hydrodynamic and environmental conditions.

Despite advancements, challenges persist in the calibration and validation of *U. prolifera* models. One of the key uncertainties in *U. prolifera* bloom modeling is the role of microbial interactions in bloom decline. Empirical evidence suggests that algicidal bacteria and viruses can significantly reduce *U. prolifera* biomass, but these processes are often neglected in existing models (Sun et al., 2022). Integrating microbial-driven mortality processes into numerical models may refine the representation of bloom dynamics and improve forecast reliability. Sparse monitoring data, especially in regions with limited infrastructure, constrain effective calibration and validation. Addressing uncertainties in input data and model assumptions remains critical for enhancing reliability. Rapidly changing conditions, such as those driven by climate change, require models capable of real-time adaptation (Gao



Number	Date	Distance(km)
F1	6/7	21.06
F2	6/21	38.71
F3	6/28	46.62
F4	7/6	65.02
F5	7/28	149.01
S1	5/18	27
S2	5/30	24
S3	6/9	42
S4	6/13	47
S5	6/26	32
S6	7/3	25
S7	7/14	19

Fig. 6. The error of geometric center point of *U. prolifera* area in model simulation.

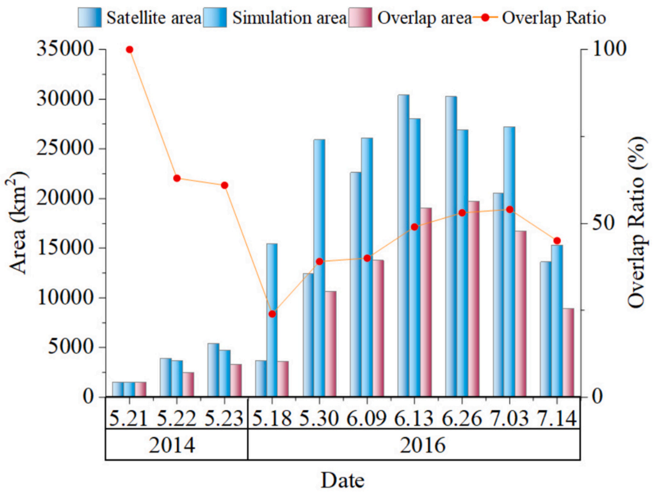


Fig. 7. Comparison between simulation results and satellite remote sensing monitoring results.

et al., 2024). To improve the calibration of biological processes in U. prolifera models, field observations of nutrient uptake rates and biomass accumulation should be integrated. Laboratory studies indicate that under high nitrogen availability, U. prolifera can achieve a maximum growth rate of 0.45 d^{-1} (Wang et al., 2022b). Model validation using in-situ nutrient concentration and biomass data will provide critical insights into the accuracy of nutrient assimilation dynamics in different bloom phases. Future efforts should focus on expanding high-resolution monitoring networks, integrating multi-source datasets, and developing hybrid models that combine mechanistic and empirical approaches. These advancements will improve the accuracy and applicability of *U. prolifera* models, contributing to better forecasting and management of green tide events.

5. Applications and prospects

5.1. Current applications of U. prolifera bloom models

Numerical models have become indispensable tools for understanding and predicting the dynamics of *U. prolifera* blooms, particularly in regions like the Yellow Sea where green tide outbreaks are frequent and impactful (Choi et al., 2023; Wu et al., 2024). Recent advancements in coupled physical-ecological models have significantly enhanced the ability to predict the growth, distribution, and decay of *U. prolifera* under varying environmental conditions (Zhou et al., 2021). Several models, such as the lagrangian transport model (LTRANS) and a coupled physical-ecological model (LTRANS-GT), have been extensively applied to simulate *U. prolifera*'s seasonal growth patterns and the factors influencing its dispersal der the interactions between biological processes (e.g., nutrient uptake and biomass accumulation) and physical factors (e.g., water temperature, current dynamics, and nutrient concentrations) (Wang et al., 2022a; Wang et al., 2022b).

The accurate modeling of *U. prolifera* blooms holds significant implications for coastal management, ecosystem health, and environmental protection. By forecasting *U. prolifera* bloom timing, location, and intensity, numerical models help mitigate economic impacts on fisheries, tourism, and recreation while reducing ecological consequences such as hypoxia, fish kills, and biodiversity loss (Glibert et al., 2014; Satta et al., 2017). For instance, the LTRANS-GT model and the floating macroalgal growth and drift model (FMGDM) have successfully simulated the trajectory and biomass of green tides, revealing interannual variability driven by initial biomass and nutrient availability (Fig. 8) (Xu et al., 2016; Zhou et al., 2021; Wang et al., 2022a). These models have also highlighted the importance of sea surface temperature, light availability, and nutrient concentrations in regulating *U. prolifera* bloom growth and decline (Choi et al., 2023). Such insights have informed proactive management strategies, including nutrient pollution control and sustainable fisheries, to minimize the adverse effects of *U. prolifera* blooms (Wang et al., 2019; Sun et al., 2022).

Numerical models have also been employed to evaluate the efficacy of nutrient management policies, such as reductions in agricultural runoff (Jang et al., 2024). By simulating various scenarios, these models provide actionable insights into the potential outcomes of different mitigation strategies, guiding policy decisions and resource management. For instance, coupled models incorporating nutrient cycling and physical transport mechanisms have demonstrated how nutrient enrichment and hydrodynamic forces contribute to *U. prolifera* bloom initiation and persistence, offering pathways to mitigate these processes (Fan et al., 2023).

Although successful, current applications of *U. prolifera* models are largely concentrated in regions like the Bohai Sea and Yellow Sea (Wang et al., 2023; Yang et al., 2023a). Expanding the geographic scope of these models to other areas experiencing similar green tide issues could enhance the global understanding of *U. prolifera* bloom dynamics. Such efforts would promote knowledge transfer across diverse ecological and socio-economic contexts, enabling the development of targeted strategies for *U. prolifera* bloom management worldwide. Moreover, the integration of global datasets and the adoption of machine learning approaches could further refine predictions, making models more adaptable to varying environmental conditions. In summary, numerical models have transformed the study of *U. prolifera* blooms, offering valuable tools for prediction, management, and policy development. Continued advancements in model integration, geographic expansion, and methodological innovation will be critical for addressing the growing challenges posed by green tides in a changing global environment.

5.2. Challenges and future research directions

Despite significant advancements in the modeling of *U. prolifera* blooms, several challenges remain that must be addressed to improve the accuracy and predictive power of numerical models. A key limitation is data acquisition. Low-resolution remote sensing imagery and spatial gaps in field measurements hinder the calibration and validation of models, leading to uncertainties in predictions (Han et al., 2022). Moreover, many models oversimplify crucial ecological processes, such as nutrient uptake kinetics and interspecies interactions, which play a

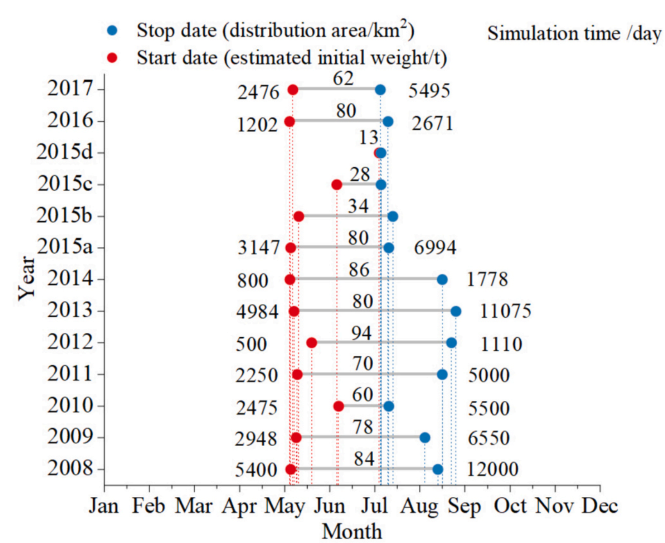


Fig. 8. Model initial locations, wet weight, start time, and simulation duration of *U. prolifera*.

significant role in *U. prolifera* bloom dynamics (Sildever et al., 2022). Overcoming these challenges will require advancements in sensor technology to capture high-resolution, multi-dimensional datasets, and a greater emphasis on interdisciplinary approaches to incorporate the complexity of biological interactions.

Another critical challenge lies in incorporating the impacts of climate change into *U. prolifera* bloom models. Rising sea surface temperatures, ocean acidification, and altered nutrient cycles are likely to affect *U. prolifera* bloom patterns in unpredictable ways (Jin et al., 2017; Sanchez-Arcos et al., 2022). Integrating these environmental changes requires substantial computational resources and innovative algorithms capable of simulating long-term climate projections. Furthermore, understanding the socio-economic impacts of green tides—such as their effects on fisheries and tourism—demands the development of coupled ecological-economic models to guide cost-effective mitigation strategies.

The complexity of ecological processes in *U. prolifera* blooms is another significant challenge. While some models, like LTRANS-GT, have successfully integrated nutrient dynamics and growth patterns, accurately simulating ecological interactions (e.g., competition, grazing, and microbial decomposition) remains difficult (Wang et al., 2022a; Wang et al., 2022b). Future research should integrate microbial ecological processes into *U. prolifera* bloom models to improve predictive capabilities. Developing coupled physical-biological-microbial models that simulate nutrient cycling, algal growth, and microbial-driven bloom decline. Utilizing machine learning approaches to analyze microbial community shifts during different bloom phases. Applying multi-omics techniques (e.g., metagenomics, metabolomics) to identify key microbial taxa influencing bloom persistence and collapse. By incorporating microbial interactions, next-generation models will provide a more holistic understanding of *U. prolifera* blooms and improve forecasting accuracy. Additionally, these models are often sensitive to variations in environmental parameters, such as wind, currents, and temperature, which can introduce uncertainties, especially during extreme weather events(Wang et al., 2022a). Therefore, future models need to address these complexities by improving the simulation of biological interactions and the representation of extreme events in model frameworks.

Spatial resolution is another area in need of improvement. Many existing models fail to capture local variations in hydrodynamic conditions and nutrient availability, particularly in complex coastal regions with artificial structures (e.g., islands and reefs) that alter local hydrodynamics (Fan et al., 2023). To improve the precision of predictions, future models must integrate higher-resolution data and improve the representation of local-scale processes.

Future research should focus on several key areas to enhance model accuracy and applicability: (1) Future models should better account for ecological interactions, including competition with other species, grazing pressure, and microbial processes influencing algae decay. This will enable models to provide a more comprehensive understanding of the long-term impacts of green tides on coastal ecosystems. (2) Incorporating higher spatial and temporal resolutions, particularly for coastal regions with complex dynamics, is crucial. The upcoming generation of Earth observation satellites offers high-resolution satellite imagery that can provide critical data for refining model inputs and predictions (Cao et al., 2023). (3) Integrating real-time oceanographic data with advanced data assimilation techniques will be essential for developing early warning systems for *U. prolifera* bloom outbreaks. This will allow for more accurate forecasts and timely mitigation efforts, enhancing coastal resource management. (4) Collaboration between modelers, ecologists, and oceanographers is necessary to enhance model accuracy. Interdisciplinary approaches that combine physical, chemical, and biological data will result in more robust models. Additionally, incorporating socio-economic factors, such as the impacts on fisheries and tourism, into integrated models will provide valuable insights for developing effective management strategies. (5) Advances in remote sensing technology, including improved satellite resolution and the use of drones, will provide better data for model calibration and real-time monitoring. Integrating data from multiple platforms—satellite, drone, and in-situ sensors—can further improve model accuracy and facilitate more effective *U. prolifera* bloom tracking.

As climate change continues to alter ocean conditions, future research must incorporate projections of temperature changes, ocean acidification, and altered nutrient cycles into forecasting models. Future research should focus on integrating nutrient assimilation models with hydrodynamic transport models to capture the full complexity of *U. prolifera* bloom formation. Advanced biogeochemical models incorporating nutrient limitation, light availability, and temperature-dependent growth kinetics will enhance predictive capabilities. Additionally, multi-omics approaches, including transcriptomics and metabolomics, can provide novel insights into the molecular mechanisms driving *U. prolifera* nutrient uptake and metabolism, leading to improved parameterization of ecological models. Models should also become more complex, integrating multi-scale spatial and temporal data to capture both small- and large-scale dynamics. Real-time updates will be critical to improving prediction accuracy. Moreover, understanding the socio-economic impacts of *U. prolifera* blooms on industries such as fisheries, tourism, and public health is essential for developing effective management responses.

6. Conclusions

Modeling and forecasting the dynamics of *U. prolifera* blooms represent a critical step in addressing the ecological and socio-economic challenges posed by green tides. The integration of biological, physical, and chemical processes within numerical models has advanced our understanding of bloom growth, dispersal, and decline. However, significant challenges remain in accurately simulating the complex interactions among environmental factors, as well as the influence of climate variability and anthropogenic activities. Improved model calibration, validation, and uncertainty analysis are essential for enhancing predictive accuracy. The development of multi-scale, coupled models that integrate real-time data, remote sensing, and socio-economic impacts will provide more comprehensive tools for managing *U. prolifera* blooms. Future research should focus on high-resolution environmental monitoring, the inclusion of climate change projections, and interdisciplinary collaborations to refine model frameworks and support effective mitigation strategies. These advancements will play a pivotal role in reducing the ecological and economic impacts of green tides, safeguarding coastal ecosystems, and informing sustainable management policies.

CRediT authorship contribution statement

Hu Chang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Ping Zuo:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization. **Yuru Yan:** Writing – review & editing. **Yutao Qin:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.marpolbul.2025.117897.

Data availability

Data will be made available on request.

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